SERA ERTUĞ



PERSONAL INFORMATION

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PROFIL

I build interdisciplinary projects that apply data and AI to engineering and the sciences. My goal is to turn research findings into real products that help people.

EDUCATION

Tevfik Fikret Science High School, İzmir (Science-Mathematics track) 2023 - 2027
Full (100%) merit scholarship

Tevfik Fikret Middle School, İzmir (Valedictorian) 2019 - 2023

Tevfik Fikret Primary School, İzmir 2015 - 2019

BİLSEM (Science and Art Center for gifted students)
Selected in primary school through a two-stage national process: a group tablet test (visual-spatial intelligence), then an individual IQ assessment. Project track in science and technology in high school.

TESTS

- AP Calculus BC (5/5)
- AP Physics 1 (5/5)
- Sir Isaac Newton Physics Contest (7th worldwide / 2,274 participants; 1st in Türkiye / 196)

LANGUAGES

- Turkish – Native
- French – Advanced (C1, DALF)
- English – Advanced (IELTS: November 2026)

INDEPENDENT RESEARCH

- Effects of extreme temperature changes in childhood on adult brain connectivity
- Fractal texture of abdominal fat as a potential biomarker for PCOS

SCIENCE & TECHNOLOGY

ETH Zurich AI Summer Week (65 selected, 23 countries) 2026
Built an AI-based business model with an international team and pitched it to a jury including Tesla and Google representatives. Honorable Mention

Beamline for Schools Competition, organized by CERN , Team Phoenix 2025
Project "Invisible Probes of Marble": muon scattering and AI-based data analysis

Junior Software Engineering Intern - VESTEL

Conrad Challenge - Conrad Innovator
Designed a tooth-mounted micro-sensor that tracks metabolic health in real time through saliva analysis

CS50x: Introduction to Computer Science (Harvard University, via edX) 2024
10 problem sets + 1 final project

Sigma STEAM Olympics - 1st Place
Cosmic Acoustics: software that turns galaxy images into music using pixel analysis and the golden ratio

Robotex International - Finalist, Software Technologies category, Team Phoenix

- International Finals, Tallinn, Estonia
- National Finals, 3rd place, Antalya, Türkiye
- Aegean Regional Finals, 2nd place, İzmir, Türkiye

Inspirit AI Scholars - Online Summer Program for High School Students
Interdisciplinary final project: AI-based sustainable agriculture

AI Marathon & Hackathon by Turkcell and the Ministry of National Education
National Hackathon Finalist, Jury Special Award

SOCIAL IMPACT

Youth Sustainable Development Leader - SÜGEP 2026

- Technical bill on photovoltaic systems at the Grand National Assembly of Türkiye
- Public awareness campaign (5,000+ signatures)
- From technical data to policy, through a field study of a LEED Platinum-certified building

Speaker and Volunteer - Global Sustainable Development Congress, Istanbul, Türkiye 2025

Youth Climate Essay Contest (Climate Cardinals & Yale Climate Connections) - Top 10 submission

Speaker at GSDC as a member of the SÜGEP team - Bangkok, Thailand 2024
Presented our project "A Model Carbon Capture Project"

ARTS & SPORTS

- ABRSM Piano Exam (Grade 5) – Passed with Merit
- Mozart Academy IV International Piano Festival – Outstanding Performance Award
- Rhythmic gymnastics (ages 6–12): national and international awards
- Tennis: tournament player